Source-Native Benchmarking for Nature-Inspired Time-Series Families: A Fail-Closed Experimental Protocol

LumenCore Technical Note - Current Public-Safe Review Draft

Abstract

LumenCore registers candidate computational families inspired by natural forms, but does not treat inspiration as evidence. This note reports a source-native benchmark ledger, the retrospective disposition of prior leads, and a frozen prospective protocol. The current ledger contains 140 registered families, 35 implementations, and 120 direct candidate-source-baseline comparisons. No candidate passes the promotion gate. The scientific contribution is therefore a reproducible comparison and falsification framework, not a performance champion.

Research Question

Can a predeclared candidate family beat every accepted baseline for a specific source, series, cadence, and horizon under prospective custody, clustered inference, an effect floor, and familywise error control?

Current Evidence Snapshot

State	Count
Registered families	**140**
Implementations present	**35**
Missing implementations	**105**
Direct candidate-source cards	**22**
Direct comparisons	**120**
Global Holm-positive comparisons	**0**
Promoted champions	**0**
Market-signal comparisons	**48**
Market comparisons inferentially insufficient	**48**
Prospective eligible observations	**0**

1. Source-Native Experimental Design

Why source-native baselines matter

A candidate can appear strong when it is compared against an inappropriate cadence, horizon, seasonal period, or baseline. This protocol therefore treats the source contract as part of the hypothesis. A family must win within each named source against the complete accepted baseline roster before any broader promotion.

Evaluation controls

- Candidate-by-source cards evaluated against the baseline roster registered for that source, cadence, series, and horizon.
- Expanding history ends immediately before each forecast origin.
- Inference clusters overlapping origins and horizons at the source-series level instead of treating every row as independent.
- Global Holm correction across the candidate-source-baseline comparison family; prior subset leads receive no preference.
- Positive, neutral, negative, inconclusive, and invalid outcomes are retained.

Registered baseline roster

- naive_last	- drift
- moving_average	- exponential_smoothing
- linear_trend	- seasonal_naive_source_period
- damped_holt_ets	- autoregressive_ridge_source_lag

Retrospective result

The former FRED and TWELVE_DATA subset leads are retired because they do not survive source-series clustering and the complete source-native baseline gauntlet. Four fixed market-signal families were also run against four registered baselines on each of Kraken, TwelveData, and AlphaVantage. All 48 market comparisons are inferentially insufficient because each source currently supplies one source-series cluster. Across the ledger, zero comparisons survive the global correction and zero candidate-source cards pass promotion.

Current conclusion: no alpha, edge, or champion is established.

2. Frozen Prospective Protocol

Candidate identity

The registry label is **fractal_brownian_surface**. The scientific estimator is **hurst_conditioned_multiscale_increment_heuristic_v1**: Hurst-conditioned weighted multiscale increment heuristic using lags 1, 2, 4, and 8 with a bounded persistence multiplier. The estimator name prevents the registry label from being mistaken for a validated fractional Brownian motion model.

Primary endpoint and multiplicity

The endpoint is relative_mean_absolute_error (rMAE = MAE_candidate / MAE_baseline). The candidate must clear 16 predeclared contrasts under one_sided_holm_familywise at familywise alpha 0.05. The maximum accepted rMAE is 0.95, corresponding to at least 5 percent relative improvement.

Future-only sample gates

- Daily arm: at least 520 future eligible sessions per series and 104 nonoverlapping five-session blocks.
- Monthly arm: at least 60 future releases per series.
- Uncertainty: synchronized_moving_block_bootstrap with 20000 frozen replications.
- Sample gates cannot be reduced after freeze.

Current protocol state

Protocol	State
LUMENCORE_TS_ SOURCE_NATIVE_20260729_V1	FROZEN_AWAITING_FUTURE_OBSERVATIONS
Promotion decision	WAITING_FOR_NEW_SOURCE_ROWS
Eligible future observations	0

Decision states

- **replicated_lead**: All three former lead contrasts must pass inside the full 16-test Holm family and each adjusted one-sided bound must clear the 5 percent effect floor.
- **source_winner**: The candidate must beat all eight source baselines, no series-horizon cell may be worse by more than 5 percent, and the p95 error ratio must not exceed 1.10.
- **cross_source_promotion**: A source winner must repeat in a second independent prospective period.
- **integrity_failure**: INVALID
- **insufficient_sample**: INCONCLUSIVE_WAITING_FOR_NEW_SOURCE_ROWS
- **primary_failure**: CORRESPONDING_CLAIM_FALSIFIED

3. Contribution, Limits, and Reproducibility

Scientific contribution

- A source-native baseline contract that prevents cross-source or cadence-mismatched promotion.
- A custody gate that binds exact source snapshots before benchmark acceptance.
- A clustered inference rule that avoids pseudoreplication from overlapping forecast origins and horizons.
- A full-family multiple-testing rule that prevents cherry-picking isolated wins.
- A future-only protocol with fixed endpoints, effect floor, sample gates, ablations, and falsification states.
- A cost-aware market-signal replay that uses identical timestamps, future-return rows, turnover costs, and source-specific baselines without granting a promotion from descriptive wins.
- A machine-readable claim boundary that keeps software proof separate from field, economic, or deployment claims.

Limitations

- 105 of 140 registered families lack implementations.
- Only 3 lanes currently have executable direct measured adapters; the wider nature-inspired registry remains inventory, synthetic stress, or context until implemented.
- All 48 market-signal comparisons are inferentially insufficient under the predeclared five-cluster minimum; descriptive score differences are not alpha or edge.
- The prospective protocol has zero eligible future observations and cannot yet support a prospective accuracy conclusion.
- No result establishes universal superiority, field performance, trading alpha, realized savings, customer acceptance, or deployment authority.

Reproducibility receipts

The ledger, prospective configuration, and prospective status are bound below. Hashes establish byte identity and custody only; they do not establish scientific correctness or external acceptance.

Artifact	Bytes	SHA-256
out/ops/source_native_family_baseline_ledger_latest.json	2147908	f4ef20f116706662dfdf09b5e62ebc9d03333c718ac38ceabb1e4eea1f301076
config/time_series_source_native_prospective_protocol_v1.json	7735	de1b3368b31244e575e473a1f7091075cff177469d97acf510d3744e6d32634b
out/ops/time_series_source_native_prospective_protocol_status.json	1108	782e563aea38144d0ce1a923475a5f4395f10f69519876df9a68ebe3eb54386a

Legacy concept-paper disposition

Three older papers are preserved as historical speculative concepts and blocked from upload. Their BioGeometry, scalar-field, bioresonance, cooling-savings, zero-point, weather-control, and wormhole-adjacent claims are not supported by the current source-native benchmark evidence.

Whitepaper payload SHA-256: c00c0483e3ab4edc3d7c0ea728446fb4b23d094b8d32ca813e42107becd7b778